

Introduction to HTML

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It defines the structure and content using a series of elements called tags. HTML is not a programming language—it's a markup language that tells browsers how to display text, images, videos, links, and other content.

Foundation of the Web

Creates the structure of every web page on the internet

Easy to Learn

Simple syntax and intuitive structure for beginners

Universal Support

Supported by all modern web browsers

Works with CSS & JavaScript

Integrates seamlessly for styling and interactivity

HTML Document Structure & Boilerplate

Essential HTML Structure

Every HTML document follows a predefined structure that helps browsers understand and display content correctly. The basic structure includes the document type declaration, root element, head section, and body section.

Complete HTML Boilerplate

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
    content="width=device-width,
    initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>Your content here</p>
</body>
</html>
```

Key Boilerplate Attributes

- **lang="en":** Specifies language for SEO and accessibility
- **charset="UTF-8":** Supports international characters and emojis
- **viewport:** Makes webpages responsive on mobile devices
- **title:** Appears in browser tabs and search results

01

<!DOCTYPE html>

Declares HTML5 document type—must be the first line

02

<html>

Root element containing all webpage content

03

<head>

Contains metadata, title, CSS files, and page information

04

<body>

Contains all visible content—text, images, videos, buttons

HTML Elements, Attributes & Content Organization

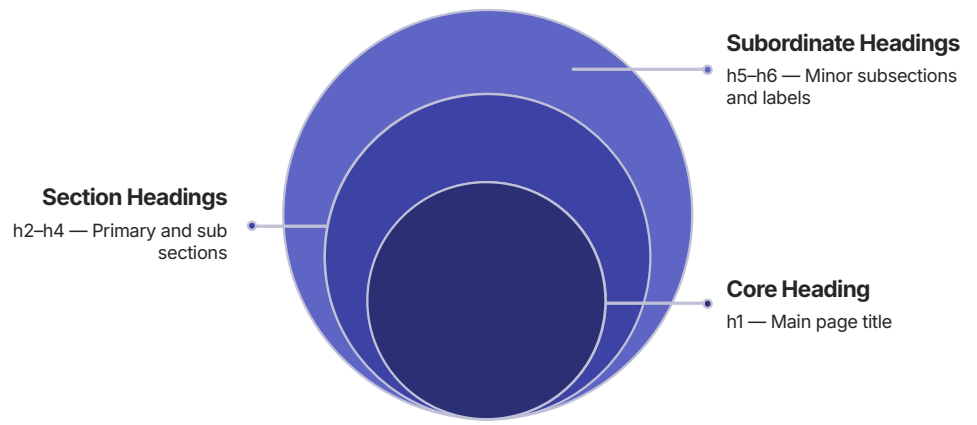
Understanding HTML Elements

An HTML element consists of an opening tag, content, and closing tag. Elements can be container elements (with closing tags) or empty elements (void elements without closing tags).

Common HTML Attributes

- **id:** Unique identifier for an element
- **class:** Groups multiple elements for styling
- **src:** Specifies image or resource location
- **href:** Defines link destination
- **alt:** Alternative text for images (accessibility & SEO)
- **title:** Displays tooltip on hover
- **target:** Controls where links open (_blank, _self)

Headings & Text Formatting



HTML provides six heading levels to create document structure. Always use only one h1 per page, follow sequential order, and use headings for structure—not just size.

Paragraphs & Line Formatting

Container Elements

`<p>Content</p>`

Have opening and closing tags

Empty Elements

`<img> <br> <hr>`

No closing tag required

Nested Elements

Elements inside other elements create parent-child relationships

- **<p> Paragraphs**

  Creates text blocks with automatic spacing. Ignores multiple spaces and line breaks in source code.

- **
 Line Break**

  Moves content to the next line without creating a new paragraph.

- **<hr> Horizontal Rule**

  Creates a horizontal separator line to divide content sections.